ALKENE

1. (c) Both 1,2 and 1,4 addition products

Electrophilic addition reaction of bromine to 1,3-butadiene proceeds via a resonance-stabilized allylic carbocation intermediate.

Step 1: Formation of carbocation

- Br_2 adds to one double bond forming a bromonium ion/carbocation intermediate.

Step 2: Resonance stabilization

- The positive charge is delocalized over the conjugated system:

$$CH_2 = CH - CH = CH_2 \rightarrow CH_2^+ - CH = CH - CH_2Br \leftrightarrow CH_2Br - CH = CH - CH_2^+$$

Step 3: Nucleophilic attack (Br^-)

- Attack at different positions gives:
  - 1,2-addition product (kinetic product)
  - 1,4-addition product (thermodynamic product)

Important Concept:

- Low temperature → 1,2-product (faster)
- High temperature → 1,4-product (more stable)

2. (b) $\underset{\underset{Br}{|}}{CH_2} - \underset{\underset{Br}{|}}{CH_2} + Zn \rightarrow \underset{\text{Alkene}}{CH_2 = CH_2} + Br_2$

3. (a) $CH_2 = CH_2 + Br_2 \xrightarrow{CCl_4} \underset{\text{1, 2-dibromo ethane}}{\underset{\underset{Br}{|}}{CH_2} - \underset{\underset{Br}{|}}{CH_2}}$

4. (b) $CH_2 = CH_2 \xrightarrow{HOCl} \underset{\underset{Cl}{|}}{CH_2} - \underset{\underset{OH}{|}}{CH_2} \xrightarrow{aq\ NaHCO_3} \underset{\underset{CH_2 - OH}{|}}{CH_2 - OH}$ (Glycol)

5. (a) Alkenes are unsaturated hydrocarbon having double bond so generally gives addition reaction.

6. (d) According to markownikoff's rule.

7. (a) $CH_3 - CH_2 - OH \xrightarrow[160-170°C]{Conc.\ H_2SO_4} CH_2 = CH_2 + H_2O$

8. (d) $$\begin{matrix} CH_3 \diagdown & & \diagup CH_3 \\ & \diagup C = C \diagdown & \\ CH_3 & & CH_3 \end{matrix} \xrightarrow[KOH\ (heat)]{KMnO_4} \underset{\text{Acetone}}{\begin{matrix} CH_3 \diagdown \\ \diagup C = O \\ CH_3 \end{matrix}} + \underset{\text{Acetone}}{\begin{matrix} O = C \diagdown \diagup CH_3 \\ CH_3 \end{matrix}}$$

9. (b) Baeyer's test

Purple Potassium permanganate ($KMnO_4$) decolourizes with alkenes due to oxidation to diol (MnO_2 forms). ALSO Used to detect Unsaturation.

10. (d) $C_2H_5I + alc.KOH \rightarrow C_2H_4 + KI + H_2O$

11. (d) $CH_2=CH_2$

Alcoholic KOH causes Elimination reaction, removing H and Cl from ethyl chloride to form ethene ($CH_2=CH_2$).

12. (a) $CH_3 - CH_2 - Br + KOH \rightarrow CH_2 = CH_2 + KBr + H_2O$

13. (a) Methane can not be obtained by Sabatier and Sendern's reaction because in this the product obtained contain minimum two carbon atoms.

$$CH_2 = CH_2 + H_2 \xrightarrow{Ni} CH_3 - CH_3$$

$$CH \equiv CH + 2H_2 \xrightarrow{Ni} CH_3 - CH_3$$

14. (c) $$CH_3 - \underset{OH}{\underset{|}{CH}} - \overset{CH_3}{\overset{|}{\underset{CH_3}{\underset{|}{C}}}} - CH_3 \xrightarrow{H_2SO_4}$$

$$\underset{2^o \text{ carbonium ion}}{CH_3 - \overset{+}{C}H - \overset{CH_3}{\overset{|}{\underset{CH_3}{\underset{|}{C}}}} - CH_3} + OH^- \xrightarrow[\text{Rearrangem ent}]{\text{Methyl shift}} \underset{\substack{\text{2,3 - dimethyl - 2 - Butene} \\ \text{[Most synnetrial Structure]}}}{CH_3 - \overset{CH_3}{\overset{|}{C}} = \overset{CH_3}{\overset{|}{C}} - CH_3} \xleftarrow[-H^+]{} \underset{3^o \text{ Carbonium ion (More stable)}}{CH_3 - \overset{CH_3}{\overset{|}{CH}} - \overset{CH_3}{\overset{|}{\underset{+}{C}}} - CH_3}$$

15. (b) ($CH_3–CH^+–CH_3$)

Carbocation intermediate forms (peroxide effect does NOT work with HCl).

16 (a) Ethylene glycol

Cold alkaline Potassium permanganate gives diol.

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18 (d) Both (a) and (b)

Forms Lewisite (β-chlorovinyl dichloroarsine).

19 (b) 2-butene

Symmetrical alkene gives 2 identical CH_3CHO.

20. (c) $CH_3-CH=CH-CH_3+HBr \xrightarrow{\text{Peroxide}} \underset{\text{2-Bromobutane}}{CH_3-CH_2-\underset{\underset{Br}{|}}{C}H-CH_3}$

Anti-markownikoff's rule is not applicable to symmetrical alkenes.

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